ALGORITHM – HYBRID INHERITANCE

Inheritance Structure: A → B → (C, D)

1. Start.
2. Create class A with variables **a = 100**, **b = 10** and method **sum()** to calculate a + b.
3. Create class B that extends A. Set **a = 10**, **b = 50** and create **sub()** to calculate b – a.
4. Create class C that extends B. Set **a = 25**, **b = 25** and create **mul()** to calculate a × b.
5. Create class D that extends B. Set **a = 100**, **b = 10** and create **div()** to calculate a ÷ b.
6. In the main() method, create an object **tv** of class D.
7. Call **tv.sub()**. Calculate 50 – 10 and display **40**.
8. Call **tv.div()**. Calculate 100 ÷ 10 and display **10**.
9. Display the results: **subtraction is 40** and **division is 10**.
10. Stop.

Quick Flow

START → Create A → Create B extends A → Create C & D extends B → Create D object → sub() → div() → Display output → STOP

Output

subtraction is 40
division is 10

Mnemonic: A → B → (C, D) = Hybrid Inheritance